



**ASIR**

# **HARDWARE FUNDAMENTALS**

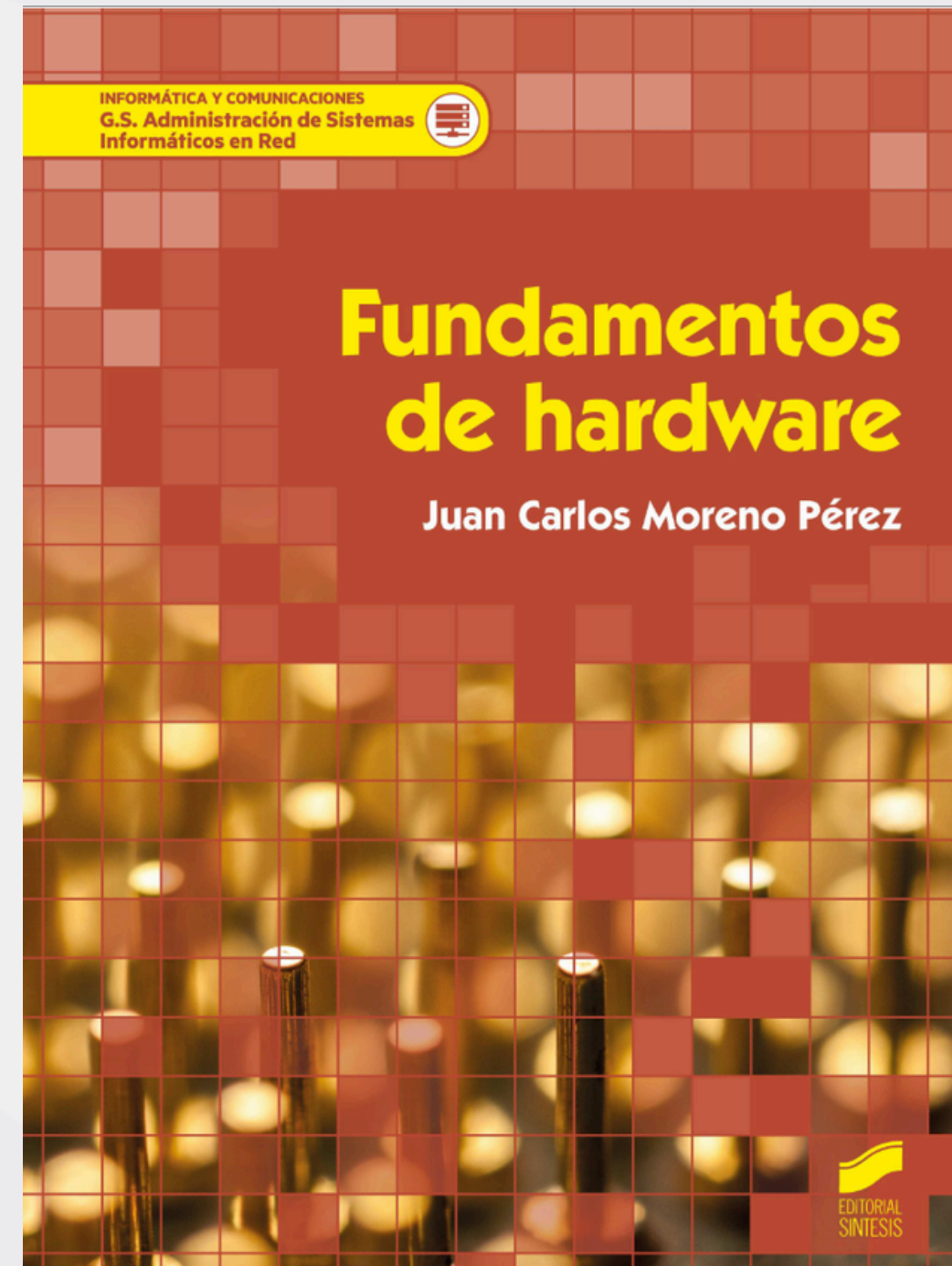
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**Teacher:**

Begoña Rodríguez Ferreras

**[begona.rodfer.1@educa.jcyl.es](mailto:begona.rodfer.1@educa.jcyl.es)**

# BOOK

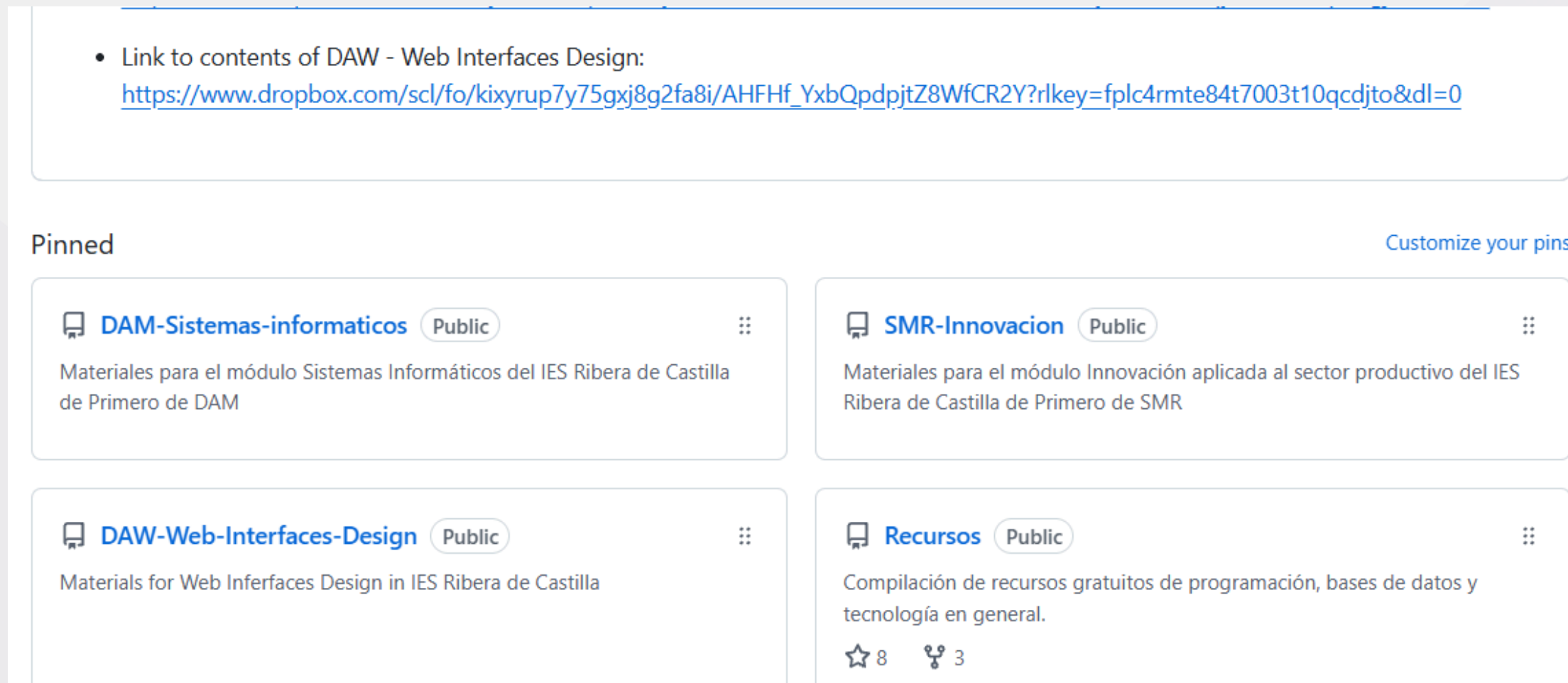


# MATERIALS

GitHub: <https://github.com/BegoRodriguez/DAW-Web-Interfaces-Design>

Dropbox:

[https://www.dropbox.com/scl/fo/kixyrup7y75gxj8g2fa8i/AHFHf\\_YxbQpdpjtZ8WfCR2Y?rlkey=fplc4rmte84t7003t10qcdjto&dl=0](https://www.dropbox.com/scl/fo/kixyrup7y75gxj8g2fa8i/AHFHf_YxbQpdpjtZ8WfCR2Y?rlkey=fplc4rmte84t7003t10qcdjto&dl=0)



The screenshot shows a user's profile page with a section titled "Pinned". At the top of this section is a link to the contents of DAW - Web Interfaces Design. Below the link are four pinned items, each with a folder icon, a title, a "Public" status, and a description. The items are: "DAM-Sistemas-informaticos", "SMR-Innovacion", "DAW-Web-Interfaces-Design", and "Recursos". The "Recursos" item also shows 8 stars and 3 forks.

- Link to contents of DAW - Web Interfaces Design:  
[https://www.dropbox.com/scl/fo/kixyrup7y75gxj8g2fa8i/AHFHf\\_YxbQpdpjtZ8WfCR2Y?rlkey=fplc4rmte84t7003t10qcdjto&dl=0](https://www.dropbox.com/scl/fo/kixyrup7y75gxj8g2fa8i/AHFHf_YxbQpdpjtZ8WfCR2Y?rlkey=fplc4rmte84t7003t10qcdjto&dl=0)

Pinned [Customize your pins](#)

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>DAM-Sistemas-informaticos</b> <span>Public</span> <br>Materiales para el módulo Sistemas Informáticos del IES Ribera de Castilla de Primero de DAM |  <b>SMR-Innovacion</b> <span>Public</span> <br>Materiales para el módulo Innovación aplicada al sector productivo del IES Ribera de Castilla de Primero de SMR |
|  <b>DAW-Web-Interfaces-Design</b> <span>Public</span> <br>Materials for Web Inferfaces Design in IES Ribera de Castilla                                |  <b>Recursos</b> <span>Public</span> <br>Compilación de recursos gratuitos de programación, bases de datos y tecnología en general.<br>★ 8    🍴 3              |

# TEAMS

I have created a classroom team for communication and project and activities delivery on Teams.

This is the code:

```
icp0asz
```

# COMPUTER ARCHITECTURES: FUNCTIONAL ELEMENTS OF A COMPUTER

## Objectives:

- Explore computing from the perspective of computer hardware and systems.
- Understand the classical Von Neumann architecture, including its functional units and subsystems.
- Learn about the architectures currently used in mobile devices.
- Understand the concept of System-on-Chip (SoC).
- Identify emerging processor architectures, including neuromorphic chips and AI-oriented architectures.

# COMPUTER ARCHITECTURES: FUNCTIONAL ELEMENTS OF A COMPUTER (II)

- Introduction
- Computer Architecture. Functional Units and Subsystems
  - Memory
  - Input/Output Subsystem
  - The ALU (Arithmetic Logic Unit)
  - The Control Unit
- SoCs. Mobile Processors
- New Microprocessor Architectures: Neuromorphic Chips
- Artificial Intelligence in Processors

# THINK ABOUT IT FOR A MOMENT...

Computers are part of everything we do. We use them without even noticing: surfing the web, writing, emailing, gaming, watching content. So let's start with a simple but powerful question: what is a computer, really? An appliance? A toy? A typewriter? Something more?



# AUTOEVALUATION

A computer is...

... a machine for writing documents

... a calculator

... a general purpose machine that works with given data and provides results

... a machine to see TV

# AUTOEVALUATION (II)

A computer is...

... a machine for writing documents

... a calculator

... **a general purpose machine that works with given data and provides results**

... a machine to see TV

# COMPUTER DEFINITION

A computer is an electronic, programmable device capable of receiving data as input, processing and storing it according to a set of instructions, and producing information as output.

A computer system and is composed of two parts: hardware (the physical elements) and software (the instructions that tell the hardware what it has to do, how to solve the problem). Sometimes, some authors include two parts more: the user and the data.

# **ACTIVITY: COMPUTER HISTORY**

Write a short article about the origins of computer science and the development of the first computers. Include references to early machines such as the Pascaline, Colossus, ENIAC, and the Intel 8086, among others.

**¿ANY QUESTION?**